MAI 5100

Fundamentals of Artificial Intelligence

Week 1: Introduction to AI

Dr. Christopher Clarke · University of Guyana · 24 October 2026

Today

- **Welcome and introductions**
- **What is AI?** Real-world impact, definitions, computational rationality, perspectives on intelligence
- **A short history of AI** and the patterns it reveals
- **AI in Guyana and the Caribbean:** opportunities, risks, and open questions
- **This course:** designing rational agents, course topics
- **Logistics:** schedule, assessment, submissions, generative AI policy

About Me

Dr. Christopher Clarke

- PhD & MS, Computer Science, U. of Michigan
- BSc, University of Guyana
- Engineering Lead, AI Core Engineering, Center for Advanced AI at **Accenture**
- Director, V75 Inc. and Nexus Hub Inc.



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My Work

Current research and engineering focus

Large-scale LLM inference

Serving large models efficiently, reliably, and at scale

Agentic systems

AI systems that plan, use tools, and take multi-step actions

Human-AI interaction

How people work with, trust, and are affected by AI

Publications: Google Scholar

Your Turn

About 60 seconds each

- Your name and background
- Why are you interested in AI?
- **Icebreaker:** What AI tool did you use this week, and did it get something wrong?

Where Are You Starting From?

- How comfortable are you writing Python?
- Have you seen probability before (conditional probability, Bayes' rule)?
- Have you used Git and GitHub?
- Have you trained or fine-tuned a machine learning model?

What is AI?

From headlines to a working definition

What is AI?

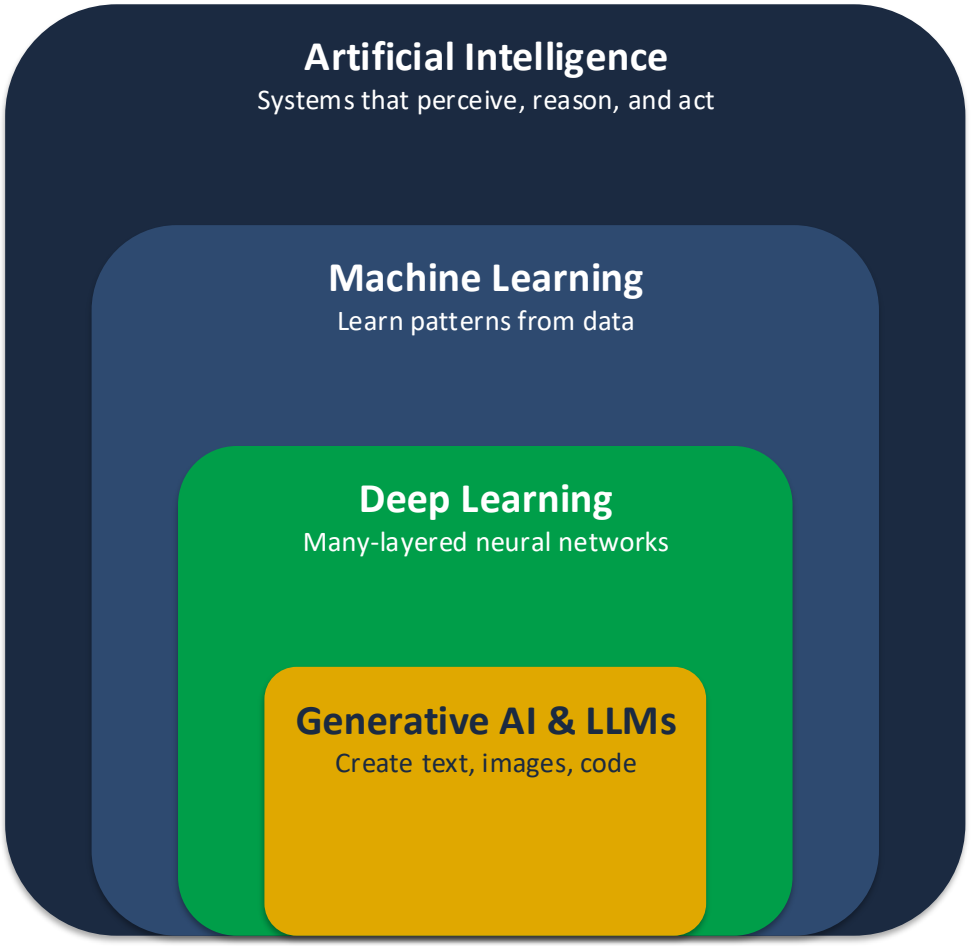
“The science and engineering of making intelligent machines.”

John McCarthy, who coined the term “artificial intelligence” (1956)

A working definition

Building computer systems that **perceive** their environment, **reason** about it, **learn** from experience, and **act** to achieve goals, doing tasks that normally require human intelligence.

- AI does not have to be **human-like**: a chess engine, a spam filter, and a chatbot are all AI
- AI is **broader than ChatGPT**: search, planning, reasoning, and learning are all part of it



In this course we take the **rational agent** view: an agent perceives its environment and acts to maximize its expected performance.

AI Is Having Real-World Impact

Everyday tools

Text assistants, search, coding, translation

Economy & labor

Investment, productivity, changing jobs

Politics & law

Regulation, elections, copyright, liability

Science

Protein structure, materials, climate, conservation

Health

Diagnosis support, triage, people seeking health advice from chatbots

Education

Tutoring, assessment, and how students learn

Social life

Companionship, content feeds, misinformation

Security

Cyberattacks, surveillance, military systems

AI Impact: Everyday Tools

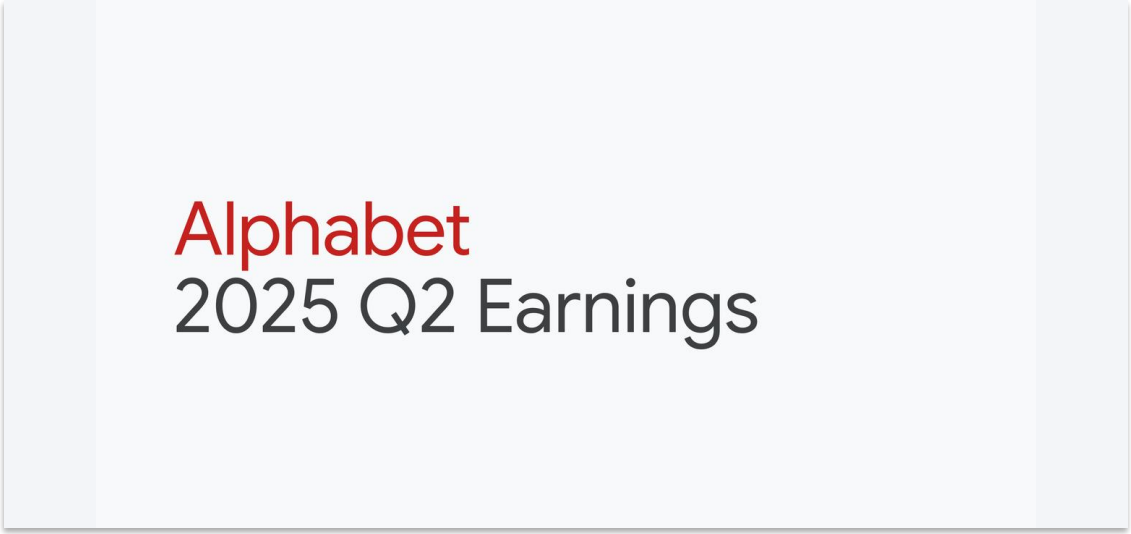


Image: Google

2B+ monthly users

Google reported that **AI Overviews** in Search had more than 2 billion monthly users (July 2025).

Source: Alphabet Q2 2025 earnings remarks



Image: Getty Images via TechCrunch

20M+ users

GitHub Copilot, an AI coding assistant, passed 20 million all-time users (July 2025).

Source: TechCrunch, Jul 30 2025

AI Impact: Economy & Labor

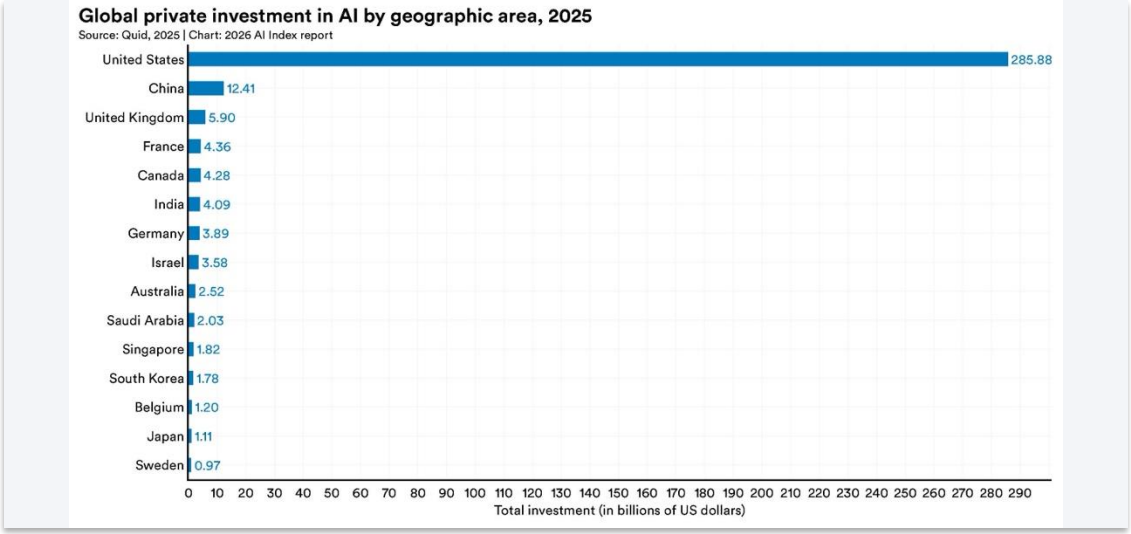


Chart: Stanford HAI, 2026 AI Index Report

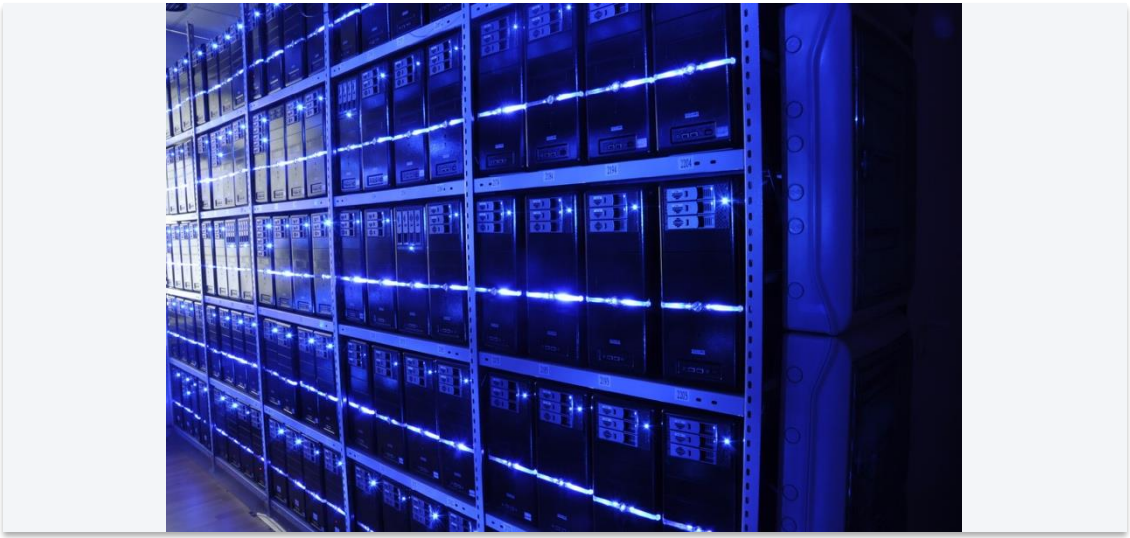


Image: BalticServers.com / Wikimedia Commons, CC BY-SA 3.0

$344.7B

Global private investment in AI in 2025, up 127.5% from 2024. The US accounted for the large majority.

Source: Stanford HAI AI Index 2026

~2 hrs/week less email

In a randomized trial of **7,137 knowledge workers** across 66 firms, workers who used Microsoft 365 Copilot spent about two fewer hours a week on email.

Source: NBER Working Paper 33795

AI Impact: Politics & Law



Image: Diliff / Wikimedia Commons, CC BY-SA 3.0

Aug 2, 2025

EU AI Act obligations began applying to providers of general-purpose AI models, including transparency and copyright requirements.

Source: European Commission



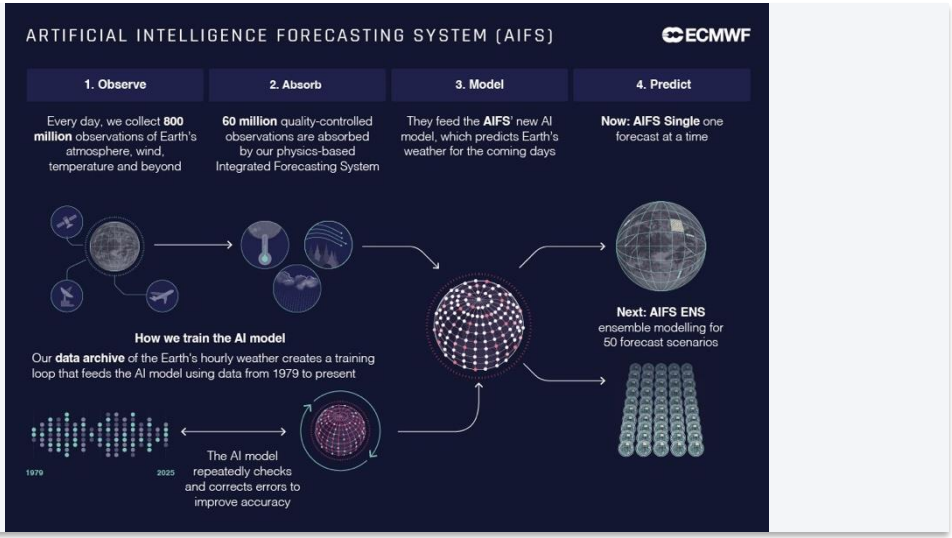
Image: H. Zell / Wikimedia Commons, CC BY-SA 3.0

7M+ pirated books

A US judge ruled that training Claude on books was **fair use**, but that Anthropic's storage of millions of pirated books was not (June 2025).

Source: Reuters, Jun 24 2025

AI Impact: Science



Graphic: ECMWF

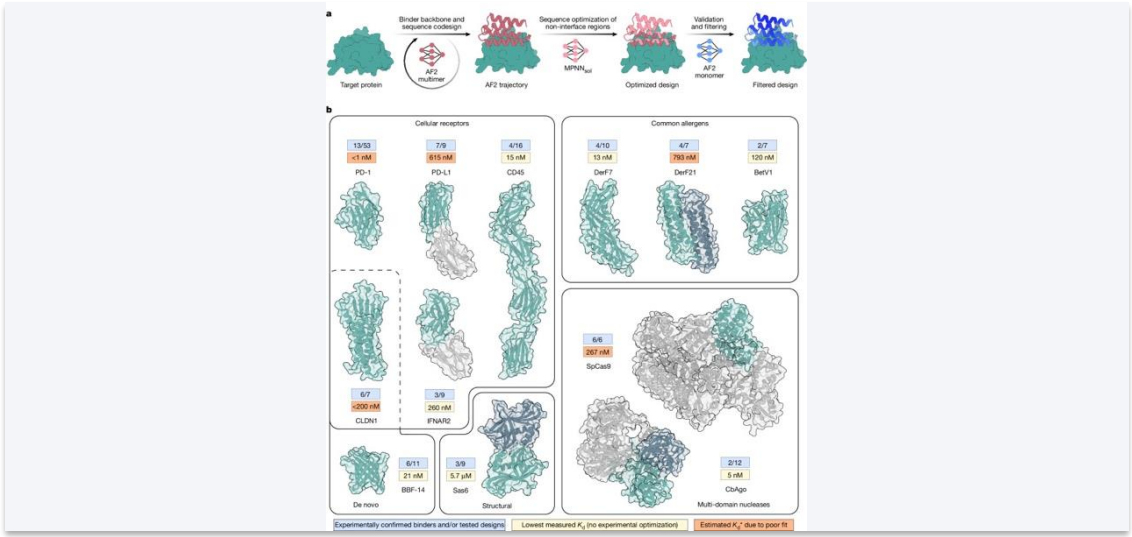


Figure: Pacesa et al., Nature (2025), CC BY

~1,000x less energy

The European weather centre's **AI forecasting system (AIFS)** became operational, cutting energy use per forecast by about 1,000 times (Feb 2025).

Source: ECMWF, Feb 25 2025

46.3% success rate

BindCraft used AI to design protein binders, averaging a 46.3% experimental success rate across 12 targets (Nature, 2025).

Source: Pacesa et al., Nature 2025

AI Impact: Health



Image: U.S. FDA

1,600+ devices

The US FDA has authorized more than 1,600 **AI-enabled medical devices** for marketing (as of Sept 2026). Authorization does not by itself prove clinical benefit.

Source: U.S. FDA



Image: National Cancer Institute / Wikimedia Commons, public domain

80.5% vs 73.8%

Sensitivity of AI-supported mammography vs. standard double reading in the **MASAI** randomized trial of ~106,000 women in Sweden, with the same specificity.

Source: Gommers et al., The Lancet, Jan 2026

AI Impact: Education

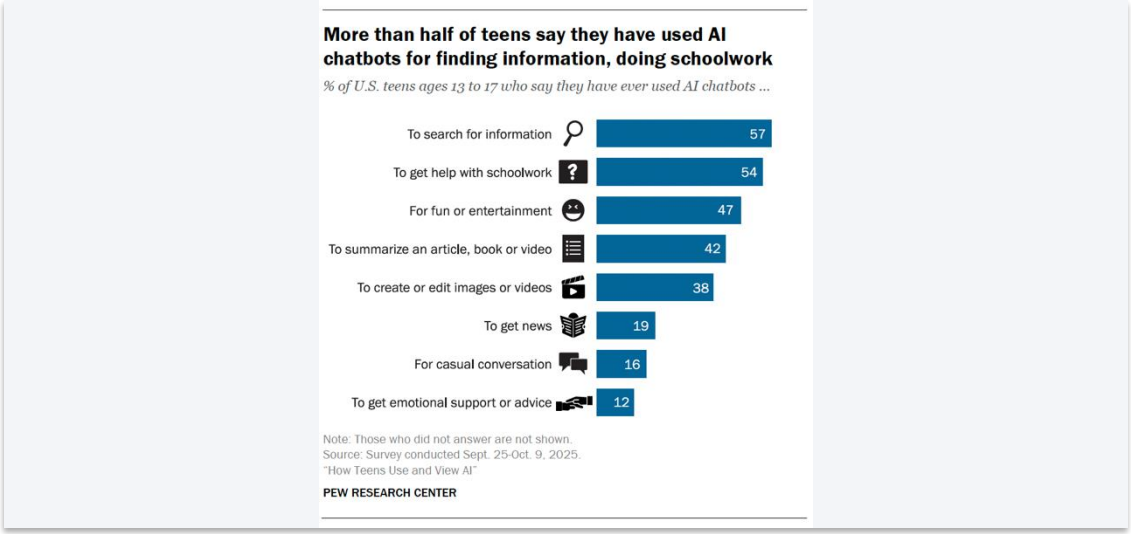


Chart: Pew Research Center

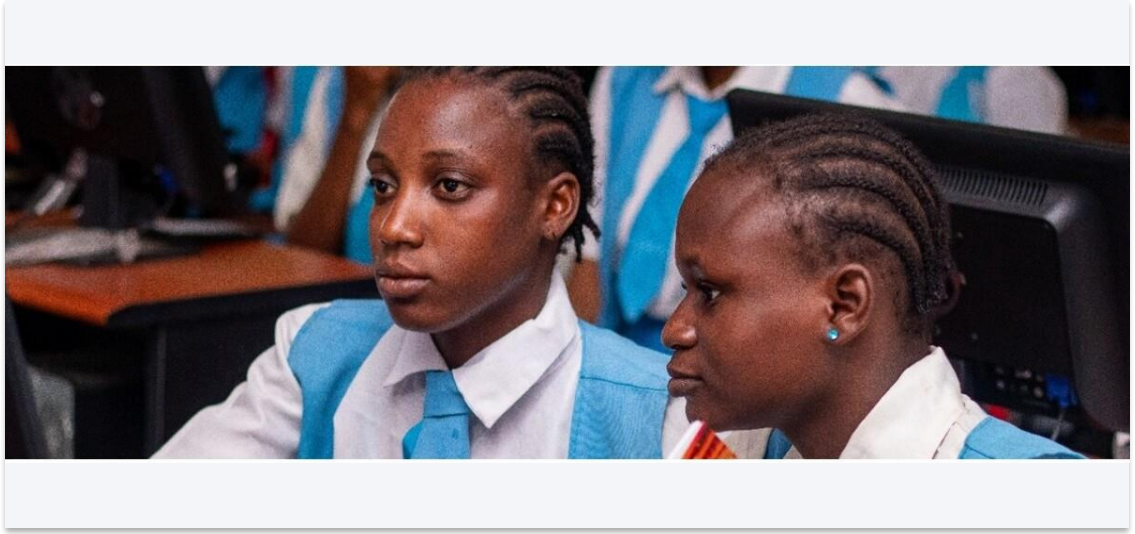


Photo: World Bank

54% of US teens

More than half of US teens say they have used AI chatbots to help with **schoolwork** (Pew survey, Sept-Oct 2025).

Source: Pew Research Center, Feb 24 2026

+0.23 SD in English

A World Bank randomized trial of a **six-week AI tutoring** program in Edo State, Nigeria, found significant gains in English learning.

Source: World Bank, May 2025

AI Impact: Social Life



Image: Common Sense Media

72% of US teens

have used an **AI companion** at least once; over half use them at least a few times a month (Common Sense Media, July 2025).

Source: Common Sense Media, Jul 16 2025



Image: AI-generated with Midjourney (2023) / Wikimedia Commons, public domain

5M views

An AI-generated video falsely claimed to show missiles striking Tel Aviv. Synthetic media like this image of the Pope spread fast and fool many people.

Source: Reuters Fact Check, Mar 10 2026

AI Impact: Security

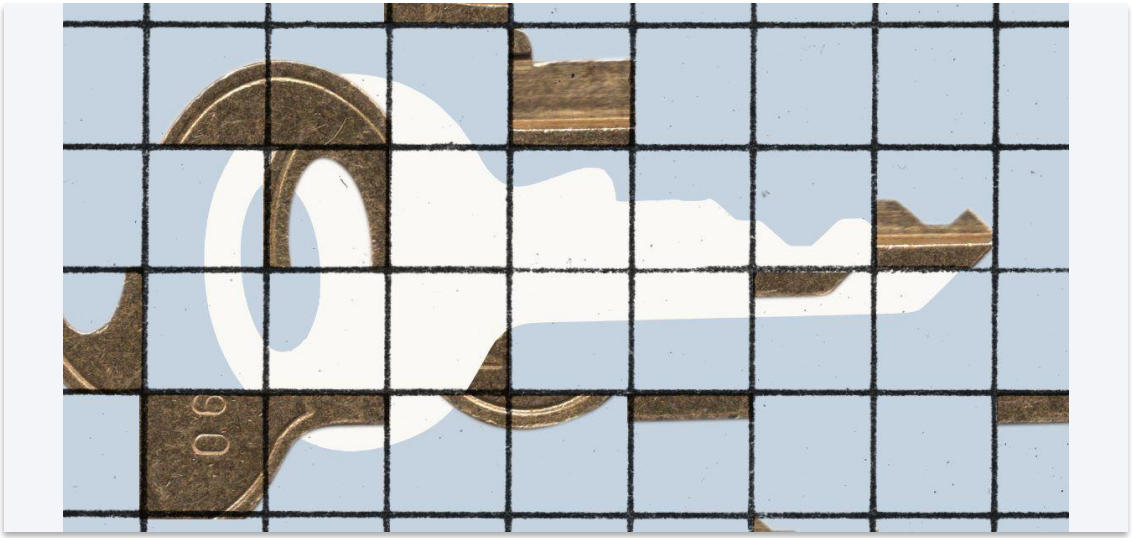


Image: Anthropic

~30 targets

Anthropic reported that a state-sponsored group used **Claude Code** to run a largely automated cyber-espionage campaign, succeeding in a small number of cases (Nov 2025).

Source: Anthropic, Nov 13 2025



Photo: ArmyInform / Wikimedia Commons, CC BY 4.0

Dozens of systems

Ukraine has rolled out dozens of **AI-assisted drone guidance** systems, some installed in thousands of drones, while requiring a human to make strike decisions.

Source: Reuters, Nov 29 2025

Discussion

Where did AI show up in your life this week?

Which of those uses would you call "intelligent"? Why?

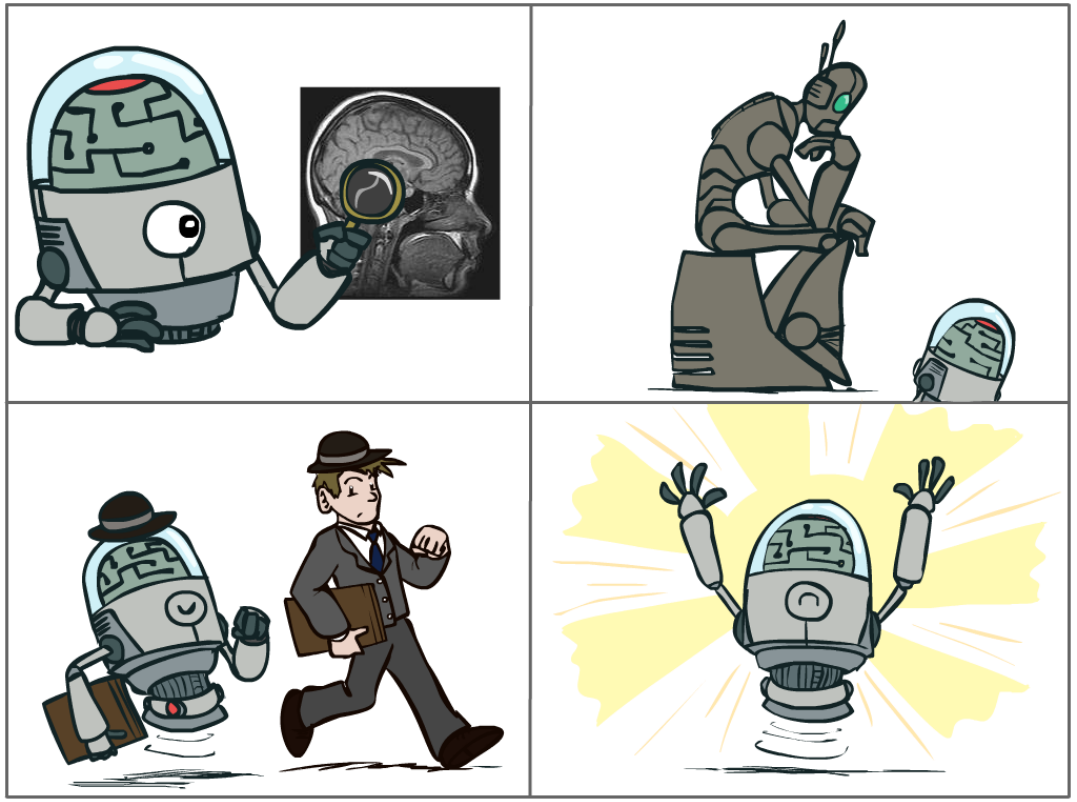
What do they have in common?

What Should We Build?

Should we make machines that...

- Think like people?
- Act like people?
- Think rationally?
- **Act rationally?**

"A computer would deserve to be called intelligent if it could deceive a human into believing that it was human." Alan Turing



Rational Decisions

- We use **rational** in a specific, technical sense:
  - Rational = maximally achieving pre-defined goals
  - Goals are expressed as the **utility** of outcomes
  - The world is uncertain, so we use **expected utility**
- **Being rational means acting to maximize your expected utility**

A better title for this course might be: **Computational Rationality**

Perspectives on Intelligence

Skills-based

"A system is intelligent only if it can do X."

Play chess? Learn from experience? Use words properly? Never make mistakes?

Embodiment (Rodney Brooks)

Intelligence emerges from a body acting in the physical world, not from abstract reasoning alone.

Psychometric (Francois Chollet)

Measure abilities, not skills: how well a system handles tasks that neither it nor its developers saw before.

Human-compatible (Stuart Russell)

1. The machine's objective is to maximize human utility
2. It is initially uncertain about human preferences
3. It learns those preferences from human behavior

Is Generality Required?

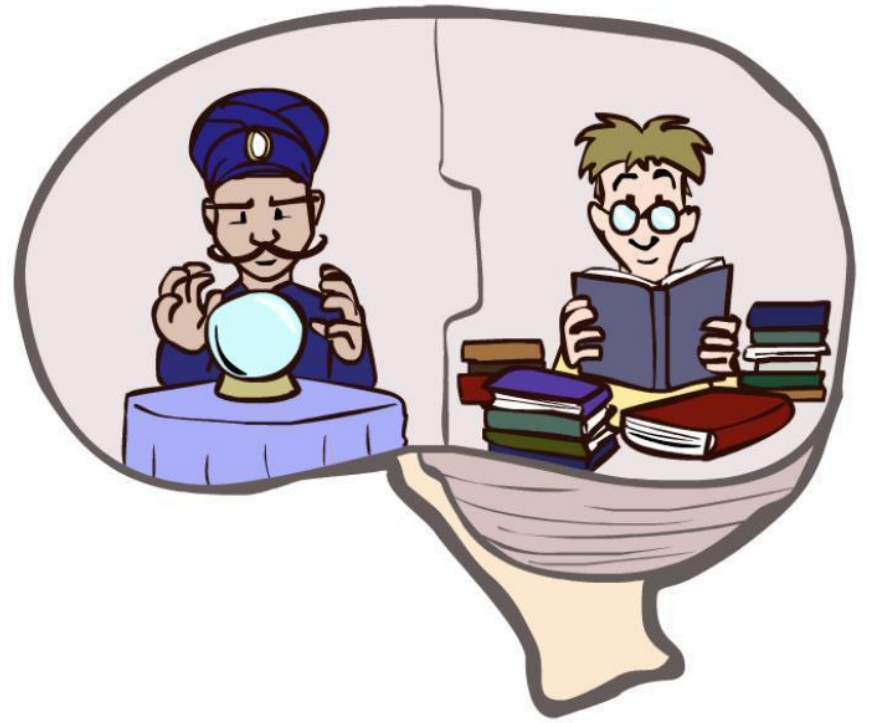
"A human being should be able to change a diaper, plan an invasion, butcher a hog, conn a ship, design a building, write a sonnet, balance accounts, build a wall, set a bone, comfort the dying, take orders, give orders, cooperate, act alone, solve equations, analyze a new problem, pitch manure, program a computer, cook a tasty meal, fight efficiently, die gallantly. Specialization is for insects."

Robert A. Heinlein

Discuss: Should AI be general like this, or are narrow, reliable systems more useful?

What About the Brain?

- Brains are very good at making rational decisions, but not perfect
- Brains are not modular like software, so they are hard to reverse engineer
- AI may already be better than brains at some tasks
- **"Brains are to intelligence as wings are to flight"**
- Scale is still different: roughly 100 trillion synapses in a human brain vs. an estimated ~1.8 trillion parameters in GPT-4
- The brain remains a great source of inspiration



Brain and model scale figures are approximate estimates.

A Short History of AI

Booms, winters, and rediscovery

1940s - 1980s: Foundations and First Booms

- **1943** McCulloch & Pitts: Boolean circuit model of the brain
- **1950** Turing, "Computing Machinery and Intelligence"
- **1956** Dartmouth workshop: the term "Artificial Intelligence" is adopted
- **1950s - 60s** Logic-driven optimism: Samuel's checkers, Newell & Simon's Logic Theorist
- **1958** Rosenblatt shows how to train a perceptron; **1969** Minsky & Papert: perceptrons cannot learn XOR
- **1970s - 80s** Knowledge-based systems and the expert systems industry boom
- **1986** Backpropagation makes multi-layer neural networks trainable
- **Late 1980s** Expert systems bust: the **AI winter**

1990 - 2017: Statistics, Data, and Deep Learning

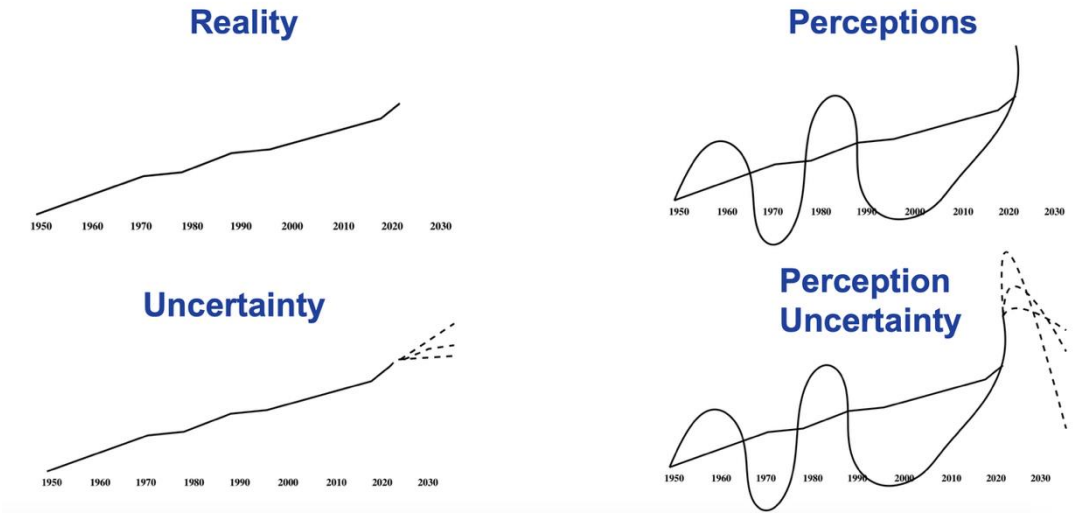
- **1990s** Resurgence of probability and a focus on uncertainty and agents
- **1992** TD-Gammon reaches human-level backgammon
- **1997** Deep Blue defeats Garry Kasparov at chess
- **2002** Roomba: AI in the home
- **2011** Watson wins Jeopardy!; Apple releases Siri
- **2012** AlexNet wins ImageNet: the deep learning era begins
- **2016** AlphaGo defeats Lee Sedol at Go
- **2017** The Transformer architecture is introduced

2017 - Today: Scale, Foundation Models, and Agents

- **2017** Libratus and DeepStack beat professionals at poker
- **2018 - 2020** AlphaFold predicts protein structure from amino acid sequences
- **2021 - 2022** Modern text-to-image generation
- **2022** ChatGPT is released; large language models go mainstream
- **2024** Nobel Prizes in Physics and Chemistry recognize AI advances
- **2025** AI systems reach gold-medal level at the International Mathematical Olympiad
- **Today** AI agents that use tools, write code, and take multi-step actions

Patterns in AI History

- **Overclaiming followed by disillusionment** (AI winters)
- **Simultaneous discovery or rediscovery** of ideas
- **Deeply interdisciplinary:** math, statistics, control, neuroscience, psychology, economics, philosophy, linguistics
- Increasingly: **law, medicine, education**
- **Where are we on the curve today?**



AI in Guyana and the Caribbean

Opportunities, risks, and the questions we must answer ourselves

Why the Local Context Matters

- Most AI systems are built elsewhere, trained on data that rarely reflects Caribbean people, places, or language
- Small states face distinct constraints: **compute cost, connectivity, data availability, talent retention**
- Climate exposure, resource economies, and dispersed populations create **problems worth solving locally**
- **Constraints can drive innovation:** efficient models, offline-first tools, and solutions designed for local needs

Regional Momentum

- **CTU ICT Week 2026** (Cayenne, French Guiana): Caribbean ICT ministers met under the theme "Advancing Resilience, Inclusion, Innovation and Cybersecurity in Caribbean Digital Development"
- Ministers **endorsed the Final Report of the Caribbean AI Task Force** and its **seven policy pillars** for regional AI governance and development
- Support for the **Regional Digital Infrastructure Plan for the Caribbean**, co-produced by the IDB and the World Bank, focused on international connectivity and data infrastructure
- Calls for **resilient telecommunications:** submarine cable diversity, early-warning systems, regional cooperation
- Trinidad and Tobago now has a **Minister of Public Administration and Artificial Intelligence**

Sources: CARICOM / CTU ICT Week 2026 Communique (caricom.org)

AI Is Already Here in Guyana

Education

The University of Guyana launched its **Master in Artificial Intelligence** in September 2024.

Public services

The Competition and Consumer Affairs Commission launched **"Damion"**, a 24-hour WhatsApp AI agent for consumer protection questions (2024).

Business

Local companies are deploying customer-facing **AI agents**, for example Silvie's Industrial Solutions' "Silvie AI" (2025).

Media

AI4D 2023 Conference, held in Guyana, showcased practical AI applications for industry and development

Companies such as V75 Inc. have been building AI systems since 2016!

Sources: Stabroek News (Mar 2024, Sep 2024, May 2025, Sep 2025)

Opportunities

Agriculture

Crop and pest monitoring, yield prediction, market information for farmers

Health

Decision support where specialists are scarce, especially in hinterland regions

Climate & disasters

Flood and hurricane forecasting, early warning, coastal monitoring

Public services

Faster, more accessible government services in local languages

Energy & resources

Optimization and monitoring across oil & gas, mining, and forestry

Education

Personalized tutoring and support at scale

Opportunity: Agriculture

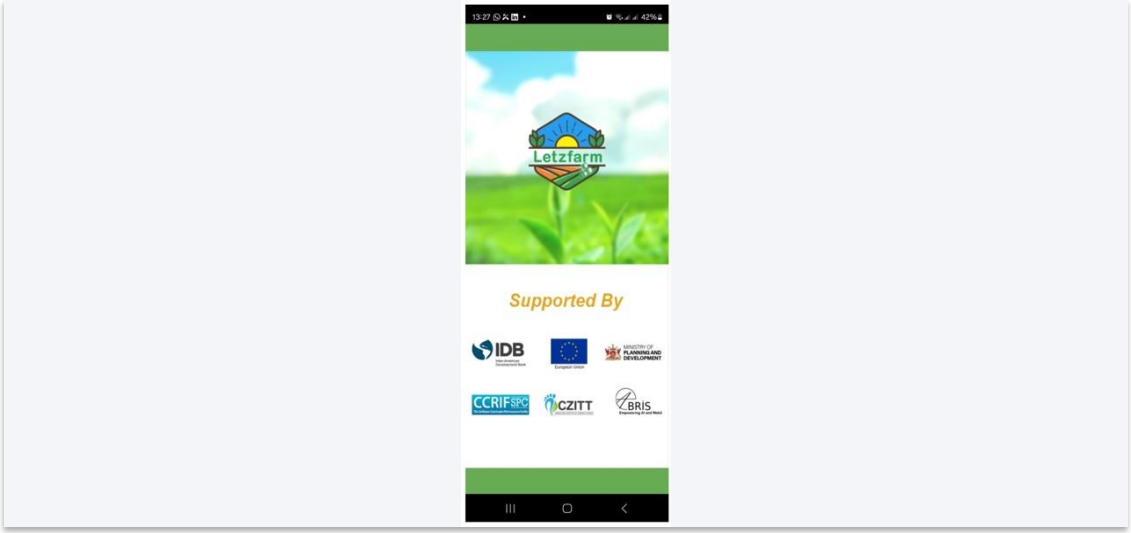


Image: CCRIF SPC

Trinidad & Tobago

LetzFarm, an AI app from the Carbon Zero Institute and Abris, helps small farmers identify pests and diseases. CCRIF granted US$25,000 to expand access (2024).

Source: CCRIF SPC



Photo: Nanditha Gogate, WELL Labs / Wikimedia Commons, CC BY-SA 4.0

10,154 advisories

India's **National Pest Surveillance System** analyses farmers' crop photos with AI, covering 61 crops and 400+ pests (as of March 2025).

Source: Govt. of India, PIB, Sep 2025

Opportunity: Health

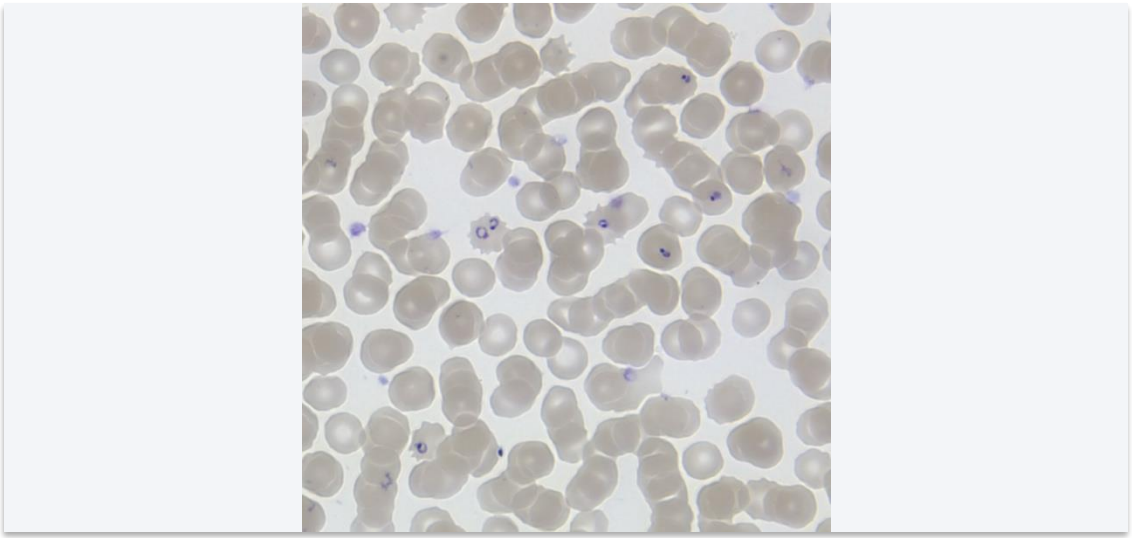


Image: malaria parasites in blood, M. Zahniser / Wikimedia Commons, public domain

Guyana: ~99% accurate

Guyana's Ministry of Health and **Harvard University** are developing an AI app that reads malaria slides. Early tests in Region Nine were about 99% accurate, per the Minister (2025).

Source: Ministry of Health Guyana, Mar 2025

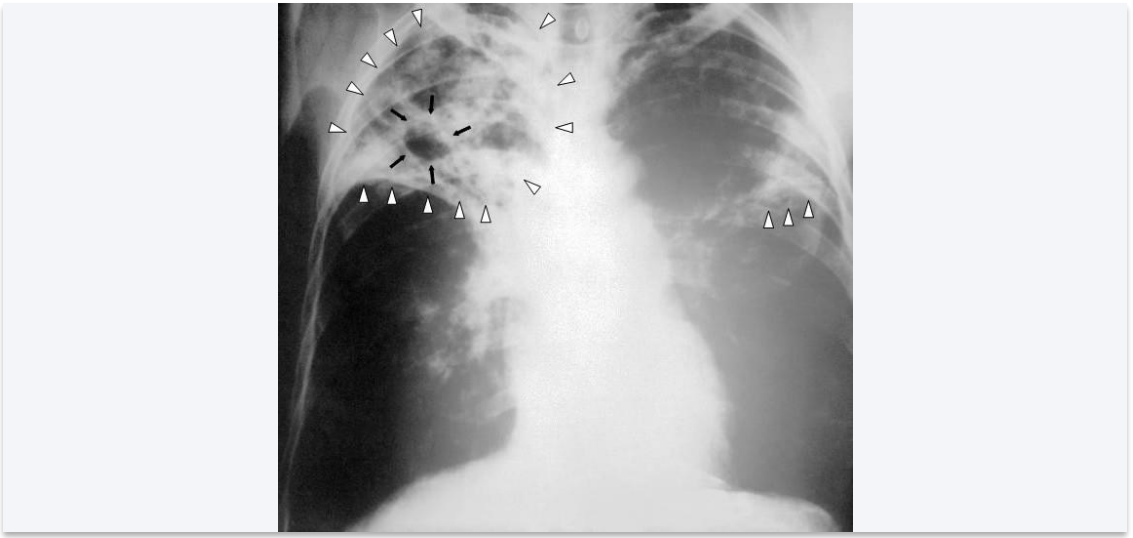


Image: Wikimedia Commons, public domain

8 states

India deployed **DeepCXR**, an AI tool that screens chest X-rays for tuberculosis, across eight states and union territories (2026).

Source: Govt. of India, PIB, Mar 2026

Opportunity: Climate & Disasters

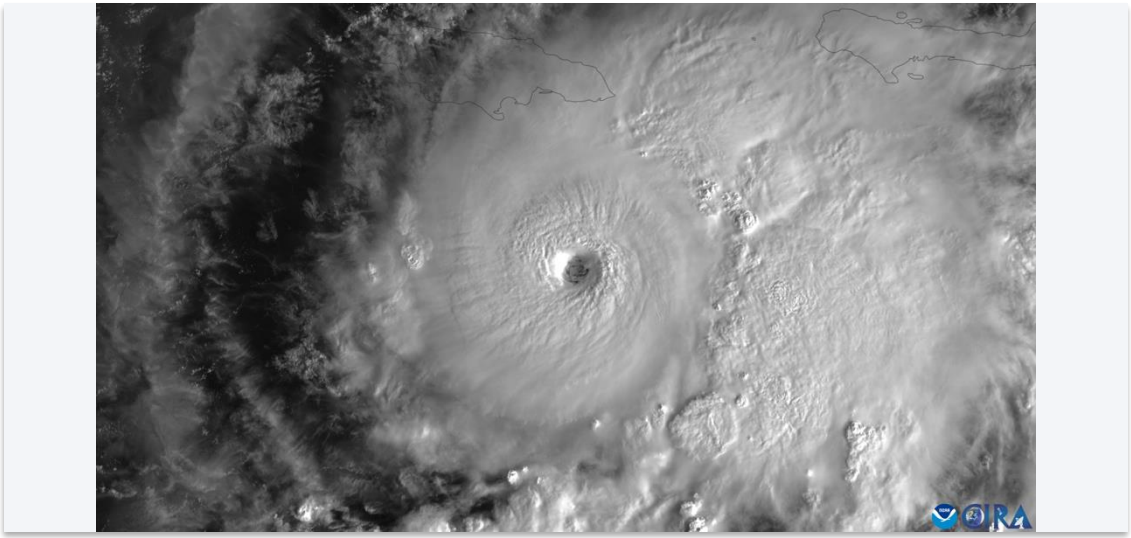


Image: Hurricane Melissa, CSU/CIRA & NOAA / Wikimedia Commons, public domain

Jamaica: 32,500 displaced

After **Hurricane Melissa**, UNDP's AI-based model estimated that up to 32,500 people may have been displaced, and satellite analysis found 4.8M+ tonnes of debris (Nov 2025).

Source: UNDP Jamaica, Nov 2025

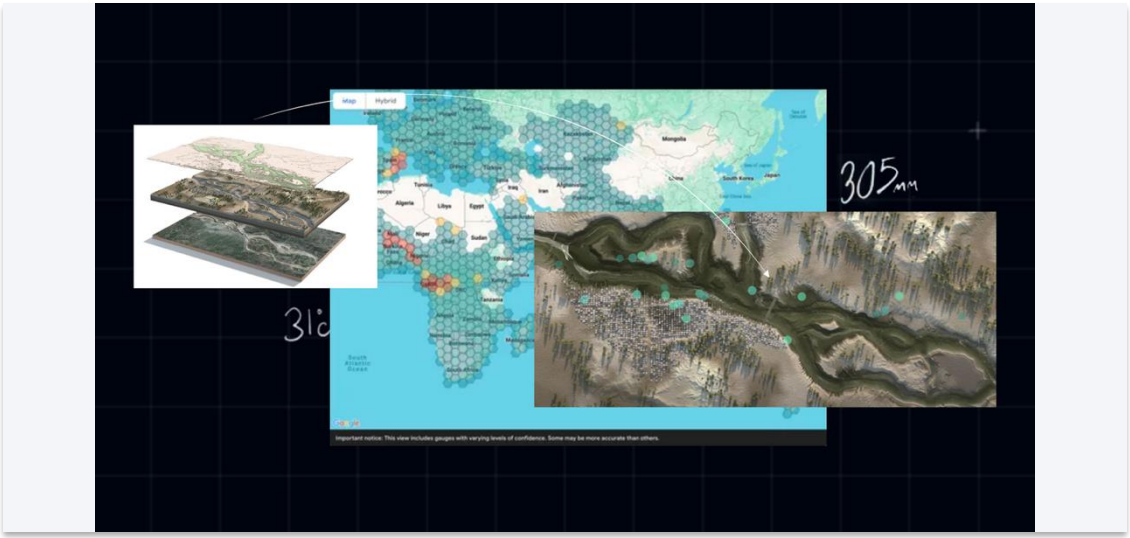


Image: Google

Cash before the flood

GiveDirectly used Google's AI **Flood Hub** forecasts to send cash to at-risk families before floods arrived, in Bangladesh and Nigeria (2025).

Source: Google, Feb 2025

Opportunity: Public Services

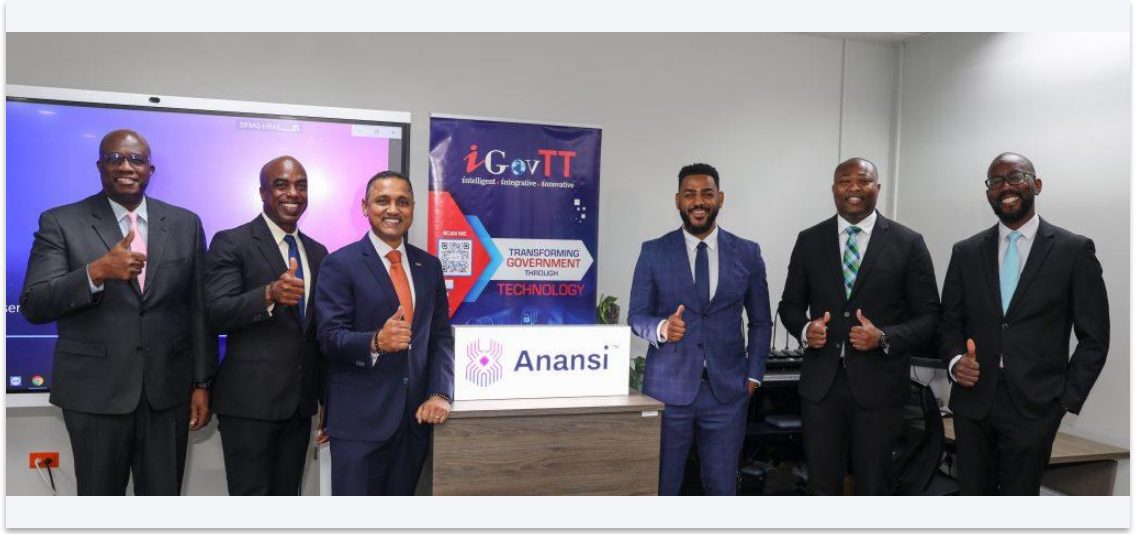


Photo: iGovTT

Trinidad & Tobago: 7,000+ FAQs

iGovTT launched **Anansi**, a 24/7 AI assistant that answers questions using FAQs from about 30 government agencies (Aug 2025).

Source: iGovTT, Aug 2025

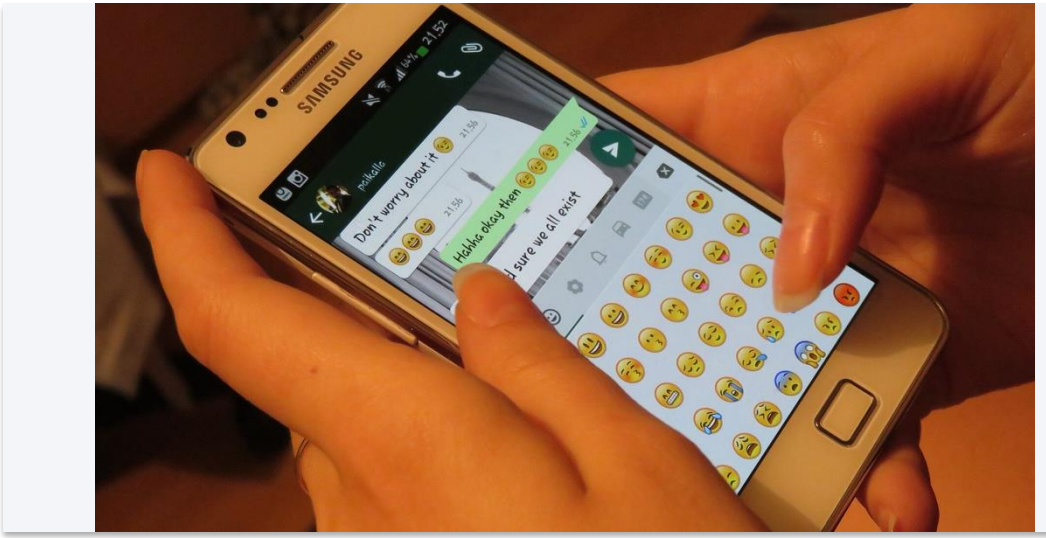


Photo: Helar Lukats / Wikimedia Commons, CC BY-SA 4.0

Brazil: Chat GOV.BR

Brazil's government answers students' questions about national exams and education programmes through an AI assistant on **WhatsApp** (2026).

Source: Govt. of Brazil, Jan 2026

Opportunity: Energy & Resources



Image: Department of Public Information, Guyana

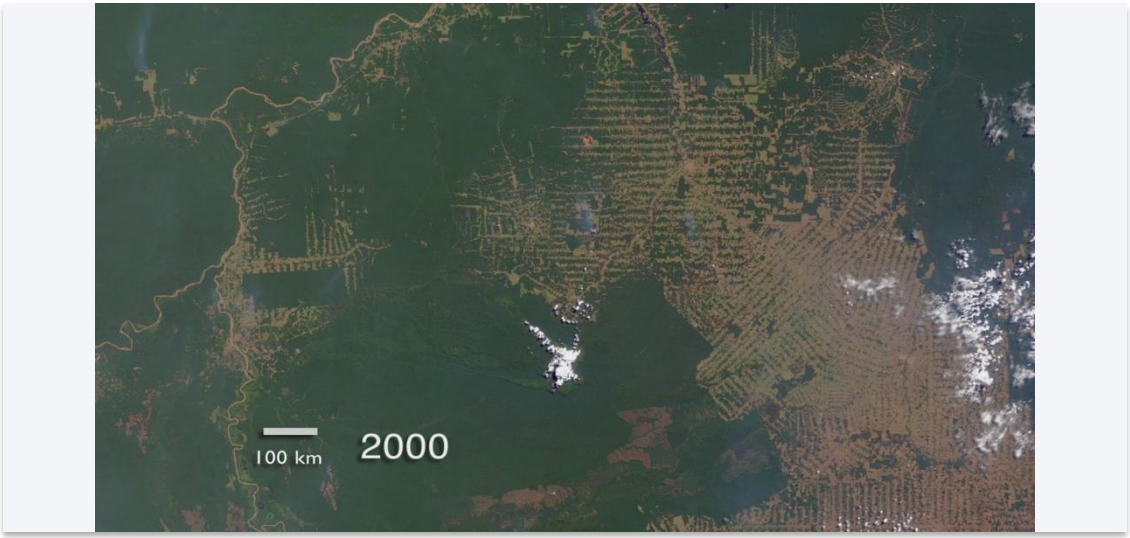


Image: NASA Goddard / Wikimedia Commons, public domain

Guyana at COP30

Guyana showcased plans to integrate **AI-enabled biodiversity monitoring** into its forest management and satellite-based forest monitoring systems (Nov 2025).

Source: DPI Guyana, Nov 2025

Brazil: 30 m maps

MapBiomas uses machine learning on satellite imagery to map land cover across Brazil every year, tracking forests, mining, and deforestation.

Source: MapBiomas, Aug 2025

Opportunity: Education



Photo: Jamaica Information Service

Jamaica: AI marking pilot

Jamaica's Ministry of Education is piloting **AI paper marking** in several schools, letting the Ministry see results in real time (Apr 2025).

Source: JIS, Apr 2025



Photo: World Bank

Nigeria: +0.23 SD

Six weeks of after-school **AI tutoring** in Edo State significantly improved English learning in a World Bank randomized trial (2025).

Source: World Bank, May 2025

Risks and Challenges

Misinformation

Deepfakes and synthetic media in elections and public discourse

Jobs

Pressure on call centres, back-office, and clerical work

Data sovereignty

Who owns, stores, and profits from Caribbean data?

Bias

Models trained elsewhere may misread local people, names, and language

Cost & access

Compute, connectivity, and licensing are expensive for small states

Talent

Skilled graduates are often recruited abroad

Risk: Misinformation



Photo: Delegation of the EU in Guyana / Wikimedia Commons

Guyana 2025 elections

EU observers reported that "AI generated videos and fake Facebook pages were misleading" and AI voiceovers were used during Guyana's 2025 general elections.

Source: EU Election Observation Mission, Final Report

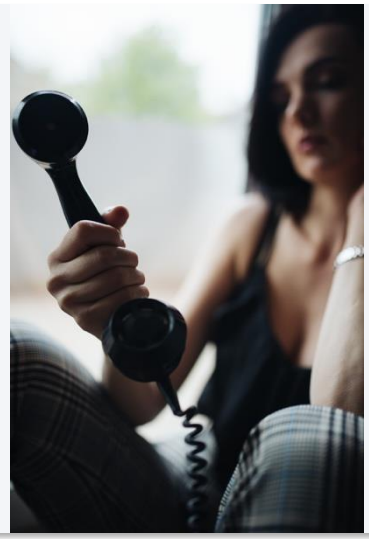


Photo: Shixart1985 / Wikimedia Commons, CC BY 2.0

US$6M fine

The US FCC fined a political consultant for robocalls using an **AI clone of President Biden's voice** telling voters not to vote (2024).

Source: US FCC, Sep 2024

Risk: Jobs



Photo: Ministry of Industry, Investment and Commerce, Jamaica

Jamaica: 50,000+ jobs

Jamaica's BPO and digital services sector employs more than 50,000 people. The government says AI is already **reshaping** customer service and back-office work (2026).

Source: MIIC Jamaica, Mar 2026

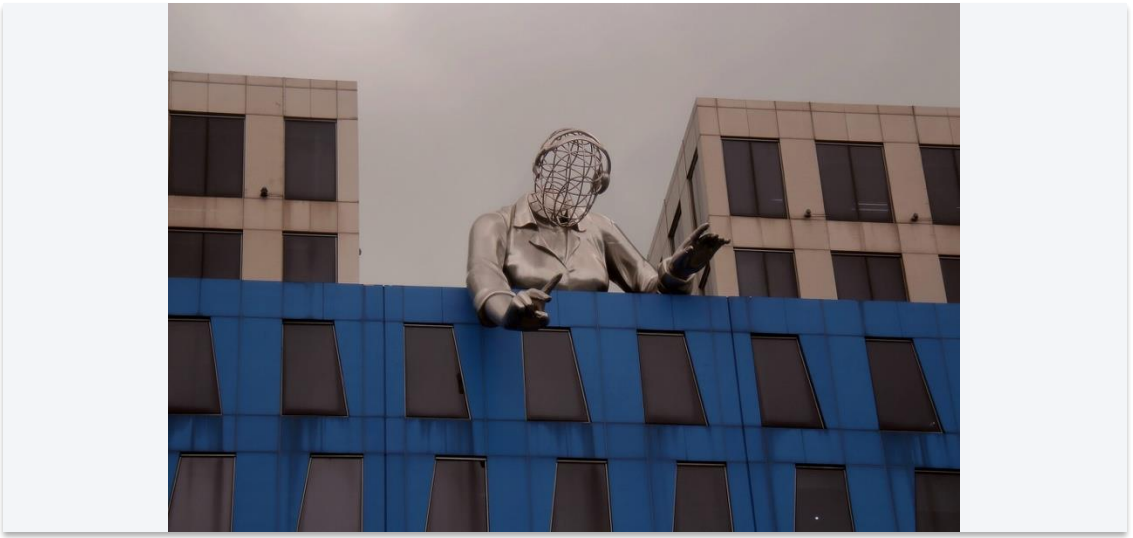


Photo: TheNuggeteer / Wikimedia Commons, CC BY-SA 4.0

Philippines: 73%

of BPO workers are contact-centre clerks, a role the IMF rates as **highly exposed** to AI with low potential for AI to complement the worker (2025).

Source: IMF Working Paper 2025/043

Risk: Data Sovereignty



Photo: David Stanley / Wikimedia Commons, CC BY 2.0

Guyana: Act No. 18 of 2023

Guyana's **Data Protection Act** sets rules for how personal data is collected, processed, and transferred. AI systems that use Guyanese data must comply.

Source: Parliament of Guyana

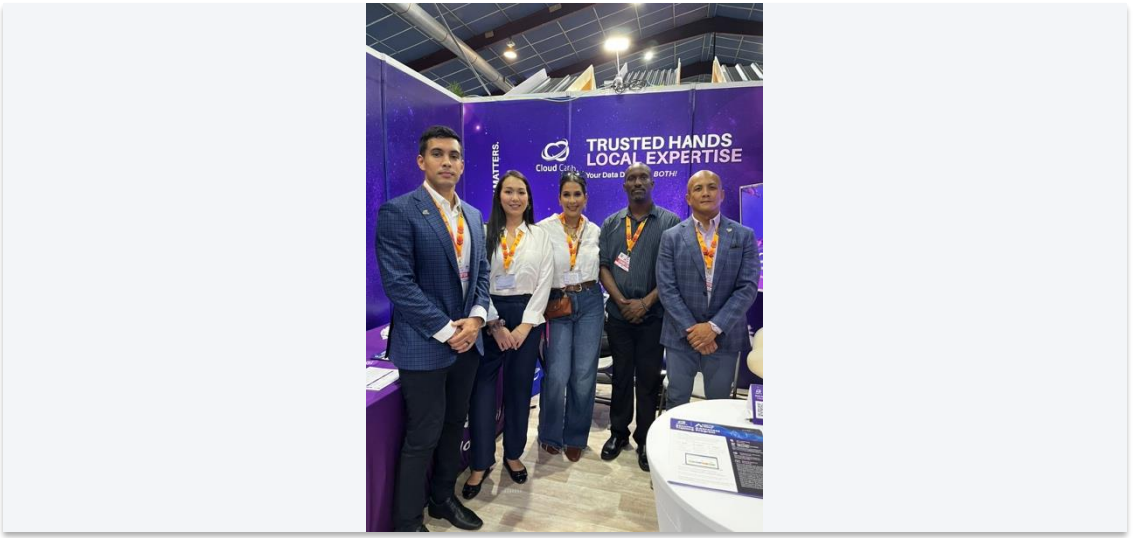


Photo: Cloud Carib

US$7M+ investment

Cloud Carib announced **sovereign data centre** pods under development in Guyana, Curaçao, and Bermuda, so regional data can stay in the region (Mar 2026).

Source: Cloud Carib, Mar 2026

Risk: Bias

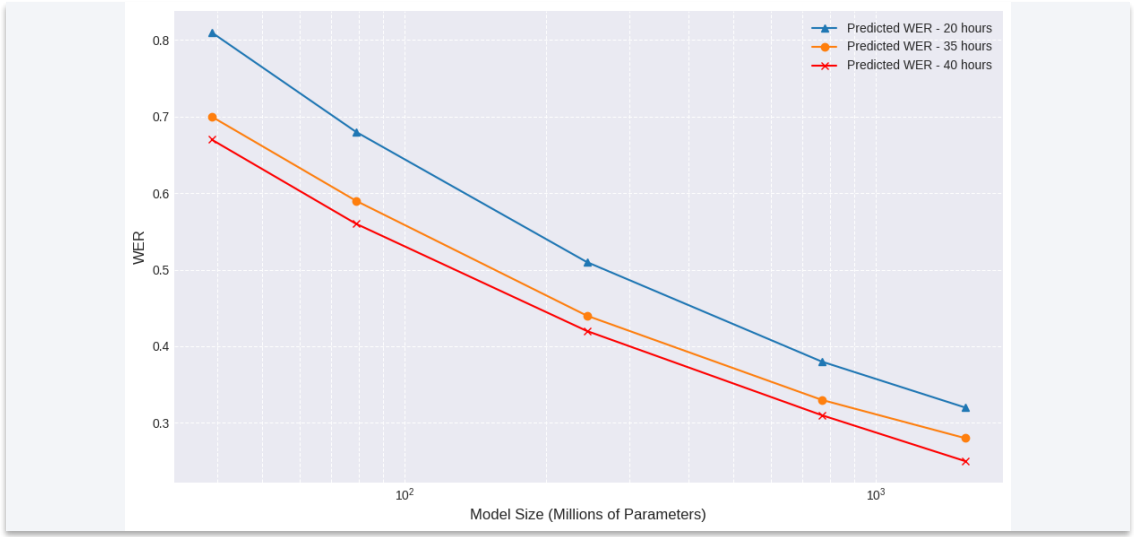


Chart: error falls with more Patois training data, from the paper (arXiv:2507.16834)

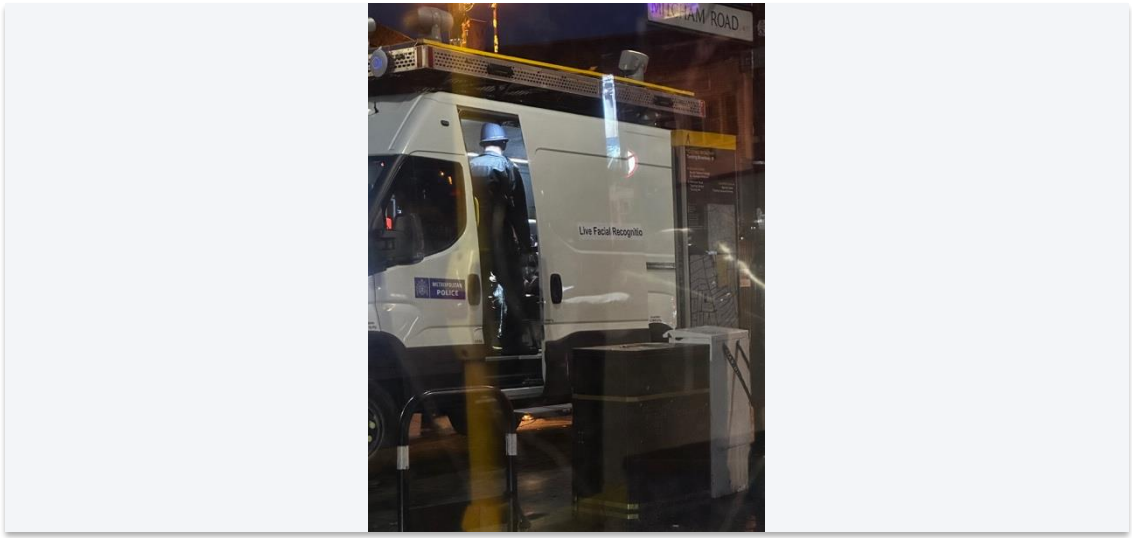


Photo: Secretlondon / Wikimedia Commons, CC BY-SA 4.0

89% vs 5% error

OpenAI's **Whisper Large** speech model had a word error rate of 0.89 on Jamaican Patois music, compared with about 0.05 on standard English (2025).

Source: arXiv:2507.16834, Jul 2025

Wrongful arrest

Detroit police arrested Porcha Woodruff, eight months pregnant, after a **faulty facial recognition match**, the sixth known false accusation from the technology (2023).

Source: ACLU, Aug 2023

Risk: Cost & Access



Photo: Stealth Communications / Wikimedia Commons, CC BY-SA 3.0

~25% of income

For the poorest fifth of households in Jamaica, fixed broadband costs nearly 25% of monthly income, versus under 5% for the richest fifth (World Bank, 2023).

Source: World Bank, Oct 2023

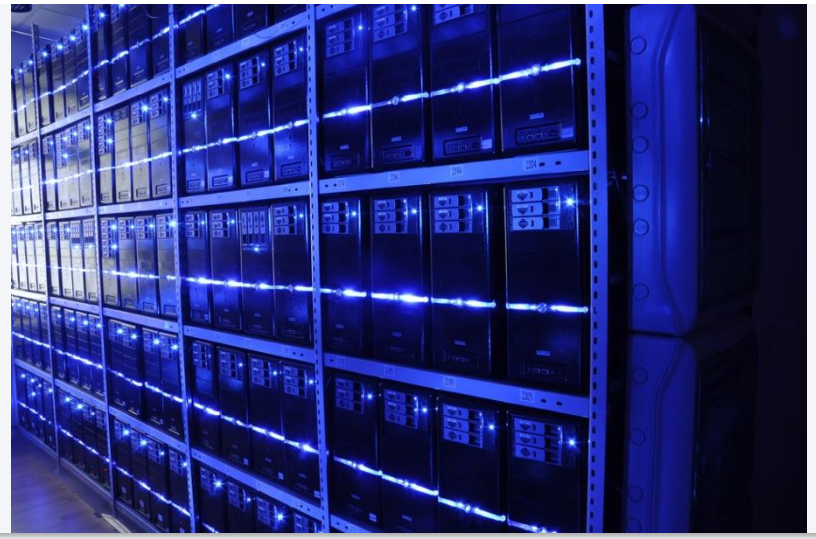


Photo: BalticServers.com / Wikimedia Commons, CC BY-SA 3.0

77% vs <0.1%

Share of global co-location **data centre capacity** in high-income vs. low-income countries (June 2025). Compute is concentrated in rich countries.

Source: World Bank, Nov 2025

Risk: Talent



Photo: AusAID / DFAT, Wikimedia Commons, CC BY 2.0

Guyana 78.4% · Jamaica 89.9%

Estimated share of Guyana-born and Jamaica-born nurses who work in **OECD countries** rather than at home. The same pressures apply to AI and tech talent.

Source: OECD, Health at a Glance: LAC 2026

Discuss

What would it take to build and keep world-class AI talent in Guyana?

Remote work for global companies? Local startups? Research funding? Partnerships with the diaspora?

Breakout Discussion (15 minutes)

Pick one problem facing Guyana or the Caribbean.

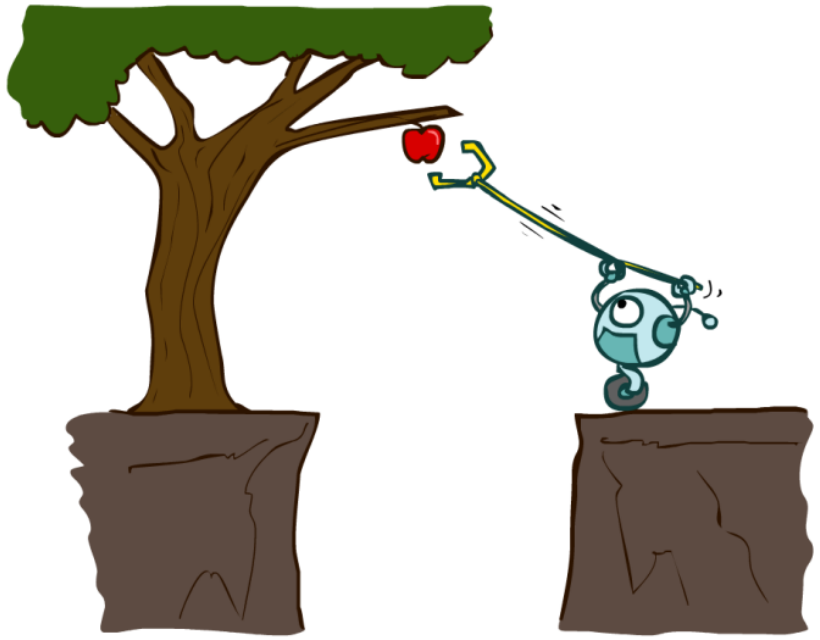
- Is AI actually the right tool, or would a simpler solution work?
- What data would you need, and does it exist?
- What would a **rational agent** for this problem be trying to maximize?
- Who benefits, and who could be harmed?
- Be ready to share your answer in two minutes

MAI 5100

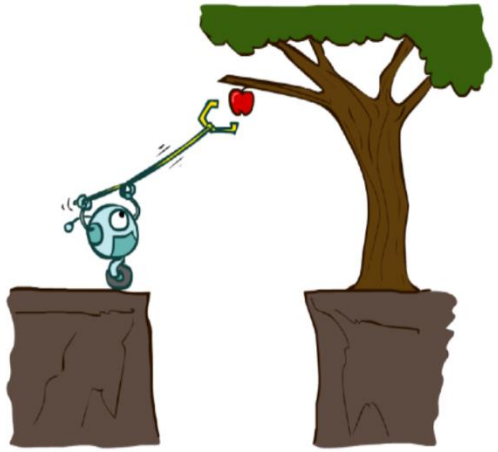
Designing rational agents

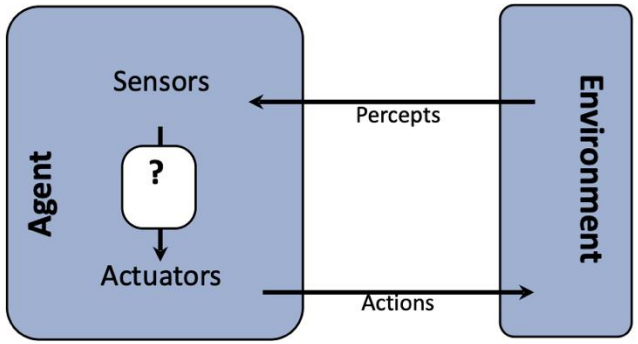
Designing Rational Agents

- An **agent** is an entity that **perceives** and **acts**
- A **rational agent** selects actions that maximize its **expected utility**
- The percepts, environment, and action space determine which techniques work
- This course covers:
  - General AI techniques for a variety of problem types
  - Recognizing when a problem can be solved with an existing technique



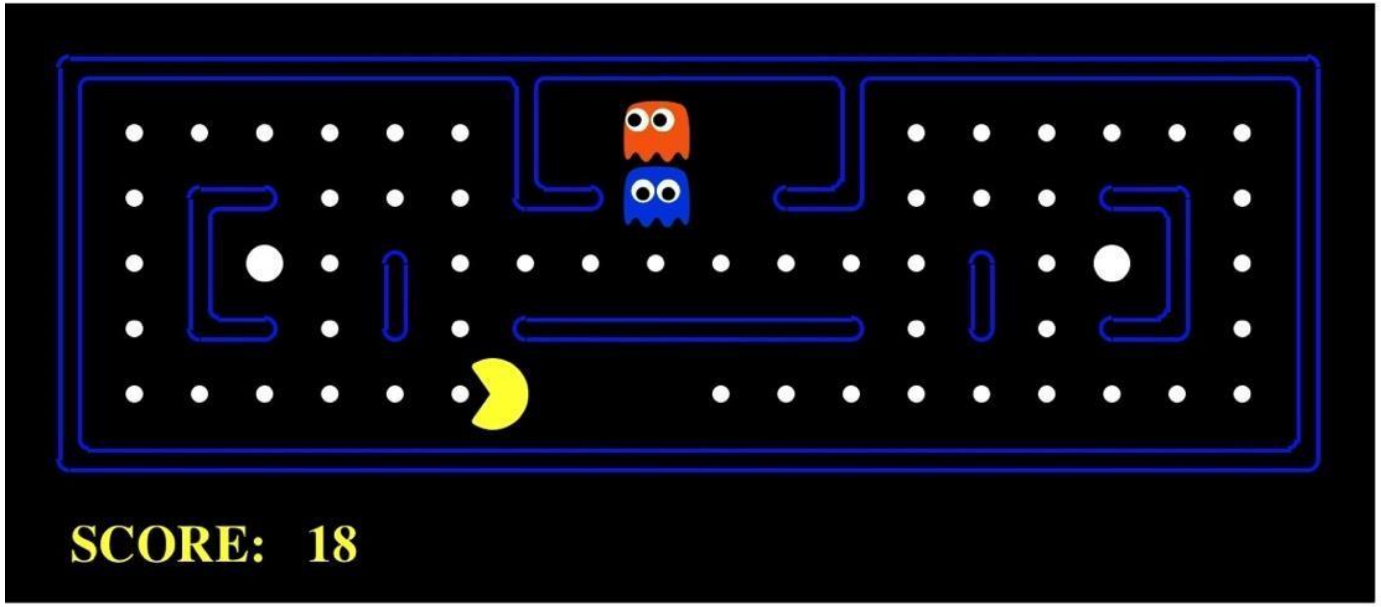
Agent and Environment





- **Sensors** receive percepts from the environment
- **Actuators** carry out actions in the environment
- The **agent function** maps what it has perceived so far to an action
- Our job: design that function well

Pac-Man as an Agent



- **Percepts:** maze layout, food, ghosts, score
- **Actions:** North, South, East, West, Stop
- **Goal:** maximize score
- You will build Pac-Man agents throughout the course

Pac-Man is a registered trademark of Bandai Namco, used here for educational purposes.

Course Topics: Core Components of Rational Agents

Search & Planning

How can I use my model of the world to find a sequence of actions that achieves my goal?

Probability & Inference

How can I make sense of uncertainty?

Reinforcement Learning

How can I learn a policy for any situation so that I maximize utility?

Machine Learning

How can I learn a model of the world from data?

By the End of This Course You Will

- Build and understand the math behind **rational, learning agents**
- **Select and apply** the right AI method for a wide range of problems
- Recognize how these methods appear in **modern AI systems**, including LLMs and agents
- Be prepared to make informed decisions about **how AI is used in society**, here and abroad

This course is a foundation, not a survey of every modern technique. The principles you learn here underpin today's state-of-the-art systems.

Course Logistics

Schedule

Week	Date	Topic
1	Oct 24	Introduction to AI
2	Oct 31	Uninformed search
	Nov 7	No class
3	Nov 14	Informed search
4	Nov 21	Constraint satisfaction
5	Nov 28	Local & adversarial search
6	Dec 5	Expectimax, utilities, MDPs
7	Dec 12	MDPs and RL
8	Dec 19	Midterm / review

Week	Date	Topic
	Dec 26 & Jan 2	No class (holidays)
9	Jan 9	Probability
10	Jan 16	Bayes nets: representation
11	Jan 23	Bayes nets: inference
12	Jan 30	Markov models and HMMs
13	Feb 6	Decision networks; ML basics
14	Feb 13	ML: classification
15	Feb 20	Project presentations

Saturdays, 9:00 AM - 1:00 PM Guyana time on Zoom. The schedule may be adjusted; changes will be announced.

Assessment and Submissions

Component	Weight
Homeworks	30%
Participation	5%
Mid-term exam	20%
Project	40%

Grades: A 80-100 · B 70-79 · C 60-69 · F < 60

Readings: Russell & Norvig, Artificial Intelligence: A Modern Approach (4th ed.)

- Homeworks are **Pac-Man based** coding projects with an autograder, plus written problems
- Submit assignments as **pull requests** to the course GitHub repository
- Due by **11:59 PM Guyana time** on the due date
- Course project: research-oriented or applied, completed in groups
- Plan for **10 - 15 hours per week** outside of class

Generative AI Policy

- You may use generative AI as a **collaborator**, and you must **acknowledge** it
- You must **understand everything you submit**; we will check this on exams
- Submitting AI output you do not understand is an academic violation
- Goal: use AI to learn faster, **not to skip the learning**

"Having GPT-4 access while solving problems significantly improves performance… However, when access is subsequently taken away, students actually perform worse than those who never had access… unfettered access to GPT-4 can harm educational outcomes."

Bastani et al., Proceedings of the National Academy of Sciences, 2025

Before Next Week

- Read **AIMA Chapter 1** (Introduction) and **Chapter 2** (Intelligent Agents)
- Make sure you have **Python 3** and **Git** installed, and a **GitHub** account
- Start thinking about **project interests** and potential teammates

Next week: Problem solving by search

Acknowledgements

- Portions of this lecture and course are adapted from **UC Berkeley CS 188: Introduction to Artificial Intelligence**
- Illustrations courtesy of the CS 188 course materials